

Intermediate Macro

Chapter 8: The Phillips Curve, the Natural Rate of Unemployment, and Inflation

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Key Tools and Concepts

- ## Assumptions

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Derivation of Phillips Curve

8.1 Inflation, Expected Inflation, and Unemployment

- In Chapter 7, we derived the following equation for wage determination:

$$W = P^e F(u, z) \quad (7.1)$$

- Impose the specific functional form $F(u, z) = 1 - \alpha u + z$ and use the price determination equation ($P = (1 + m)W$) to derive:

$$P = P^e(1 + m)(1 - \alpha u + z) \quad (8.1)$$

- The equation (8.1) can be rewritten as follows:

$$\pi = \pi^e + (m + z) - \alpha u \quad (8.2)$$

Derivation of Phillips Curve

8.1 Inflation, Expected Inflation, and Unemployment

- Introducing time subscripts, equation (8.1) becomes:

$$P_t = P_t^e(1 + m)(1 - \alpha u_t + z)$$

- Next, divide both sides by last year's price level P_{t-1} :

$$\frac{P_t}{P_{t-1}} = \frac{P_t^e}{P_{t-1}}(1 + m)(1 - \alpha u_t + z) \quad (8A.1)$$

- Take the fraction P_t/P_{t-1} on the left side and rewrite it as:

$$\frac{P_t}{P_{t-1}} = \frac{P_t - \mathbf{P_{t-1}} + \mathbf{P_{t-1}}}{P_{t-1}} = 1 + \frac{P_t - P_{t-1}}{P_{t-1}} = 1 + \pi_t$$

Derivation of Phillips Curve

8.1 Inflation, Expected Inflation, and Unemployment

- Do the same for the fraction P_t^e/P_{t-1} on the right side:

$$\frac{P_t^e}{P_{t-1}} = \frac{P_t^e - \mathbf{P}_{t-1} + \mathbf{P}_{t-1}}{P_{t-1}} = 1 + \frac{P_t^e - P_{t-1}}{P_{t-1}} = 1 + \pi_t^e$$

- Replacing P_t/P_{t-1} and P_t^e/P_{t-1} in equation (8A.1) by the expression we have just derived,

$$(1 + \pi_t) = (1 + \pi_t^e)(1 + m)(1 - \alpha u_t + z)$$

- Divide both sides by $(1 + \pi_t^e)(1 + m)$:

$$\frac{(1 + \pi_t)}{(1 + \pi_t^e)(1 + m)} = 1 - \alpha u_t + z \quad (8A.2)$$

Derivation of Phillips Curve

8.1 Inflation, Expected Inflation, and Unemployment

- So long as inflation, expectation inflation, and the markup are not too large, a good approximation of equation (8A.2) is given by

$$\pi_t - \pi_t^e - m = z - \alpha u_t$$

- Rearranging gives:

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t \quad (8.3)$$

- Dropping the time indexes, this is equation (8.2). With the time indexes, this is equation (8.3).

Understanding of Phillips Curve

8.1 Inflation, Expected Inflation, and Unemployment

- It is important to understand each of the effects at work:
 - ① An increase in expected inflation π^e leads to an increase in actual inflation π .
 - ② Given expected inflation, an increase in markup m , or an increase in the factors that affect wage determination—an increase in z —leads to an actual inflation.
 - ③ Given expected inflation, a decrease in the unemployment rate u leads to an increase in actual inflation.
- It will be convenient to use time indexes so that we can refer to variables in a specific year.

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t \quad (8.3)$$

The Original Phillips Curve

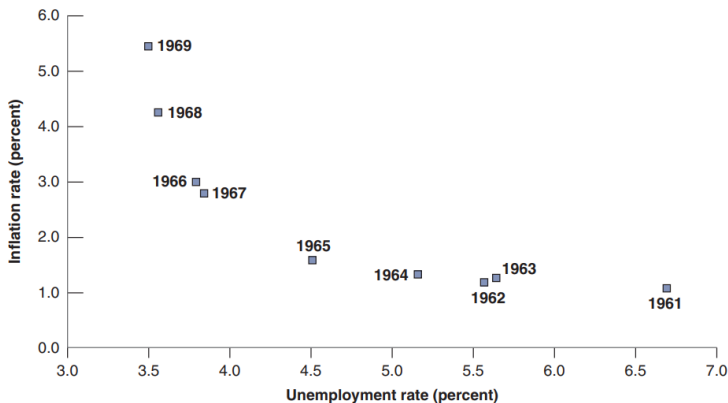
8.2 The Phillips Curve and Its Mutations

- Assume that inflation varies from year to year around some value $\bar{\pi}$ and also that inflation is not persistent, so that inflation this year is not a good predictor of inflation next year.
- In such an environment, it makes sense for wage setters in particular to expect that inflation next year will simply be equal to $\bar{\pi}$.
- In this case, $\pi^e = \bar{\pi}$ and equation (8.3) becomes:

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t \quad (8.4)$$

Inflation versus Unemployment in the United States, 1961-1969

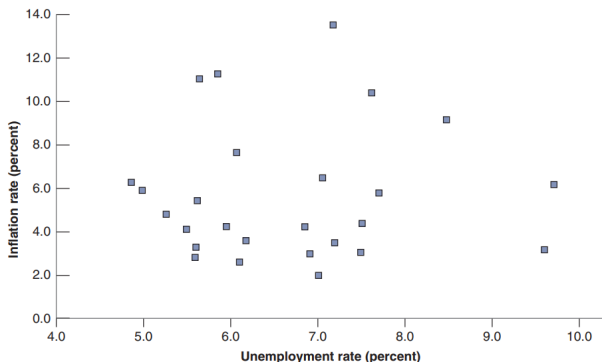
8.2 The Phillips Curve and Its Mutations



Inflation versus Unemployment in the United States, 1970-1995

8.2 The Phillips Curve and Its Mutations

- Around 1970, the negative relation between the inflation rate and the unemployment rate, so clear during 1961-1969, broke down.



The De-anchoring of Expectations

8.2 The Phillips Curve and Its Mutations

- Why did the original Phillips curve vanish?

The De-anchoring of Expectations

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8.2 The Phillips Curve and Its Mutations

- Why did the original Phillips curve vanish? *Because wage setters changed the way they formed their expectations about inflation.*
- This change came from a change in the behavior of inflation.
The rate of inflation became more persistent.
- As a result, when forming expectations, people started to take into account the persistence of past and of current inflation.
- In macroeconomic jargon, expectations that had been **anchored** became **de-anchored**.

Changes in Expectation Formation

8.2 The Phillips Curve and Its Mutations

- Now suppose expectations of inflation are formed according to:

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1} \quad (8.5)$$

- We can think of what happened in the 1970s as an increase in the value of θ over time.
- When $\theta = 1$, the relation becomes

$$\pi_t - \pi_{t-1} = (m + z) - \alpha u_t \quad (8.6)$$

- Note that now a new relation emerged, this time between the unemployment rate and the *change* in the inflation rate.

Accelerationist Phillips Curve

8.2 The Phillips Curve and Its Mutations

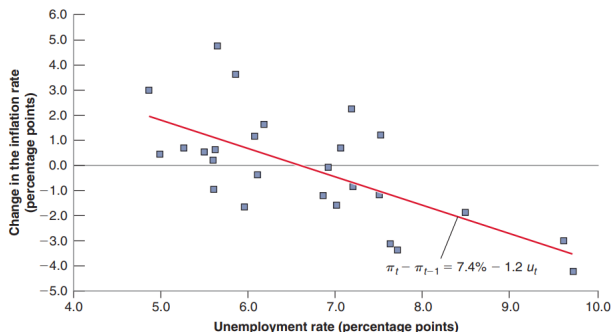
- The equation (8.6) is often called the modified Phillips curve, the augmented Phillips curve, or the **accelerationist Phillips curve**.
- Original Phillips curve: Low $u_t \Rightarrow$ High inflation
Accelerationist Phillips curve: Low $u_t \Rightarrow$ Increasing inflation
(and thus an *acceleration of the price level*)

Change in Inflation versus Unemployment in the United States, 1970-1995

8.2 The Phillips Curve and Its Mutations

- The figure plots the relation that best fits the scatter of points for that period and is given by:

$$\pi_t - \pi_{t-1} = 7.4\% - 1.2 u_t \quad (8.7)$$



The Re-anchoring of Expectations

8.2 The Phillips Curve and Its Mutations

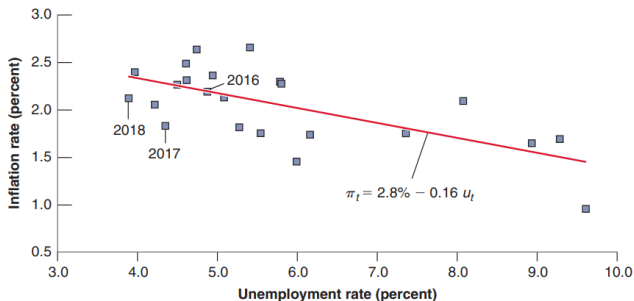
- In the 1990s, the Phillips curve relation changed yet again, because of a change in monetary policy.
- From the early 1980s on, many central banks increasingly emphasized their commitment to maintaining low and stable inflation.
- By the mid-1990s, the Fed had largely achieved its goal and so the way people formed expectations changed again.
- Expectations of inflation that had become *de-anchored* during the 1970s and the 1980s became *re-anchored*.
- In terms of equation (8.5), θ went back down to zero, and the Phillips curve returned to the original Phillips curve.

Inflation versus Unemployment in the United States, 1996-2018

8.2 The Phillips Curve and Its Mutations

- The figure plots the line that best fits the scatter of points for the period 1996-2018 and is given by:

$$\pi_t = 2.8\% - 0.16 u_t \quad (8.8)$$



Lesson Learned

8.2 The Phillips Curve and Its Mutations

- So are we back to the original Phillips curve? Yes, but having learned an important lesson.
- The Phillips curve relation is a relation among inflation, expected inflation, and unemployment. What this implies for the relation between inflation and unemployment thus depends very much on *how people form expectations*. And this in turn depends very much on the behavior of inflation.

Discovery of the Concept of u_n

8.3 The Phillips Curve and the Natural Rate of Unemployment

- The history of the Phillips curve is closely related to the discovery of the concept of the natural rate of unemployment that we introduced in the Chapter 7.
- In the late 1960s, two economists, Milton Friedman and Edmund Phelps, questioned the existence of a trade-off between unemployment and inflation.
- They questioned it on logical grounds, arguing that such a trade-off could exist only if wage setters systemically underpredicts inflation and that they were unlikely to make the same mistake forever.
- The unemployment rate could not be sustained below a certain level that they called the *natural rate of unemployment*.

Connection between the Phillips Curve and u_n

8.3 The Phillips Curve and the Natural Rate of Unemployment

- Recall that the definition of u_n is the unemployment rate at which the actual price level is equal to the expected price level.
- Imposing the condition that $\pi = \pi^e$ in equation (8.3) gives:

$$0 = (m + z) - \alpha u_n$$

- Solving for the natural rate u_n ,

$$u_n = \frac{m + z}{\alpha} \quad (8.9)$$

- Now rewrite equation (8.3) as

$$\pi_t - \pi_t^e = -\alpha \left(u_t - \frac{m + z}{\alpha} \right)$$

Implications of the Equation

8.3 The Phillips Curve and the Natural Rate of Unemployment

- Note from equation (8.9) that the fraction on the right side is equal to u_n , so we can rewrite the equation as:

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n) \quad (8.10)$$

- This is an important equation, and one you must remember: It links the inflation rate, the expected inflation rate, and the deviation of the unemployment rate from the natural rate.
- The equation also gives us a way of getting an estimate of the natural rate.

A Summary and Many Warnings

8.4 A Summary and Many Warnings

- Taking stock of what we've learned:
 - The relation between inflation and unemployment depends on how wage setters form expectations of inflation.
 - If expectations are anchored, the Phillips curve takes the form of a relation between inflation and unemployment.
 - If expectations are unanchored, the Phillips curve becomes a relation between the change in inflation and unemployment.
- In short, there is a relation between inflation and unemployment, but it is a complex one.
- Further warnings: The natural unemployment rate itself moves over time; it varies across countries; the relation between unemployment and inflation can completely disappear when inflation becomes very high; and it can also disappear when inflation becomes very low and turns into deflation.

Variations in the Natural Rate over Time

8.4 A Summary and Many Warnings

- We saw earlier that the estimate for the natural rate of unemployment for the period 1970-1995 was 6.2%, but the estimate for the period starting in 1996 was down to 5%.
- In estimating equation (8.7) or equation (8.8), we implicitly treated $m + z$ as a constant.
- But there are good reasons to believe that m and z vary over time.
- Will the natural rate of unemployment remain low in the future?
- Note that the interpretation of the changes in the natural rate tend to come after the fact. Such changes are difficult to predict.

8.4 A Summary and Many Warnings

- On average, European economies have much higher unemployment rates than does the United States.
- Some economists attribute high European unemployment to **labor market rigidities**, a term applied to a collection of policies, including generous unemployment insurance, a high degree of worker protection, bargaining rules that protect unions, and high minimum wages.
- The relationship between such policies and unemployment is not straightforward.

High Inflation and the Phillips Curve

8.4 A Summary and Many Warnings

- The relationship between inflation and unemployment may depend on the degree of inflation.
- When the inflation rate becomes high, inflation tends to become more variable. As a result, workers and firms become more reluctant to enter into labor contracts that set nominal wages for a long period of time.
- For this reason, the terms of wage agreements change with the level of inflation.
- **Wage indexation**, which is a provision that automatically increases wages in line with inflation, becomes more prevalent.
- These changes lead in turn to a stronger response of inflation to unemployment.

Wage Indexation and the Effects of Unemployment on Inflation

8.4 A Summary and Many Warnings

- Imagine an economy that has two types of labor contracts. A proportion λ of labor contracts is indexed. A proportion $(1 - \lambda)$ of labor contracts is not indexed.
- Under this assumption, equation (8.10) becomes:

$$\pi_t = [\lambda\pi_t + (1 - \lambda)\pi_t^e] - \alpha(u_t - u_n)$$

- If we assume that $\pi_t^e = \pi_{t-1}$, we get:

$$\pi_t = [\lambda\pi_t + (1 - \lambda)\pi_{t-1}] - \alpha(u_t - u_n) \quad (8.11)$$

or

$$\pi_t - \pi_{t-1} = -\frac{\alpha}{(1 - \lambda)}(u_t - u_n)$$

8.4 A Summary and Many Warnings

- The relation between unemployment and inflation can completely disappear when inflation becomes low and turns into deflation.
- The Phillips curve relation between inflation and unemployment can disappear when inflation is near zero.

